Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

50% task-completion time horizon doubles every ~7 months — from 2 seconds (GPT-2, 2019) to ~2 hours (o3, 2025)

Source: METR — arXiv:2503.14499 (March 2025)

A New Metric: 50% Task-Completion Time Horizon

1

Assemble Task Suite

Create diverse suite of 170 software/research tasks ranging from 3 seconds to 8 hours

2

Measure Timing & Success

Time skilled humans; measure AI agent success rates on each task

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Find task duration where AI achieves 50% success rate

The 50% time horizon provides an intuitive, human-grounded measure of AI capability — allowing comparison across vastly different models from GPT-2 (2 sec) to Claude 3.7 Sonnet (54 min).

Fig. 2: Methodology overview

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST
97 tasks
1 min – 30 hrs

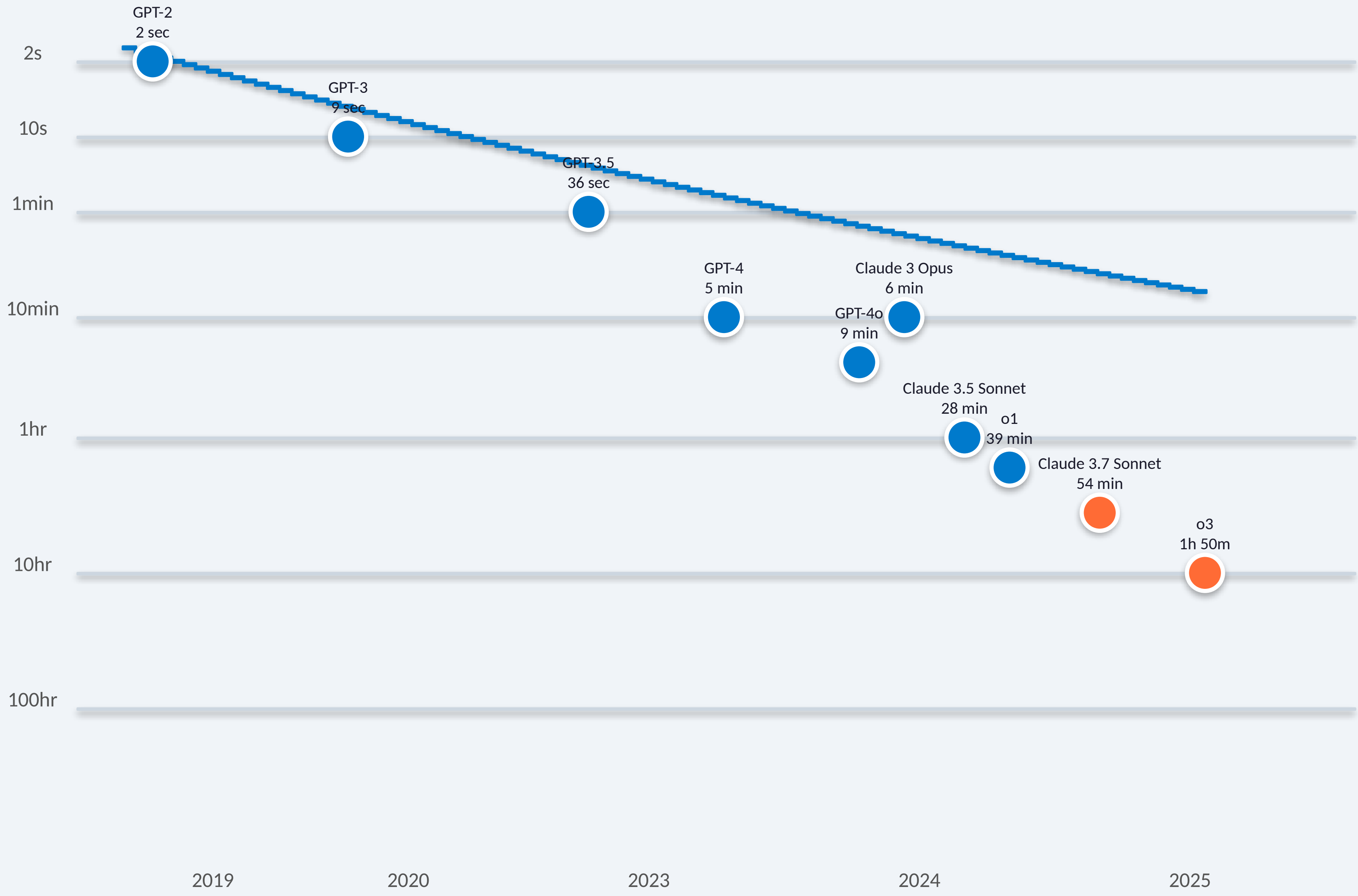
RE-Bench
7 tasks
All 8 hours

SWAA
66 tasks
1 sec – 30 sec

800+ human baselines collected totaling 2,529 hours of human timing data

Task	Duration	Example Task Description
find_shell_script	3 sec	Multiple choice: Which file is a shell script?
wikipedia_research	1 min	Research simple factual information from Wikipedia
oxdna_simple	9 min	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 min	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hrs	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon: 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years



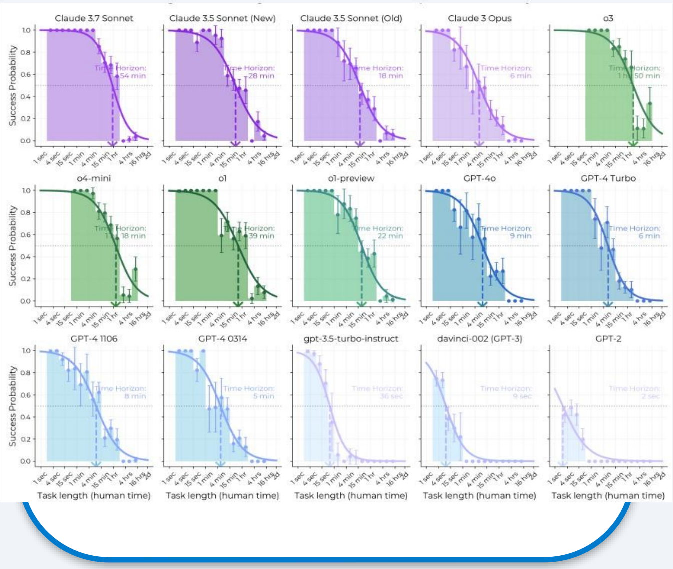
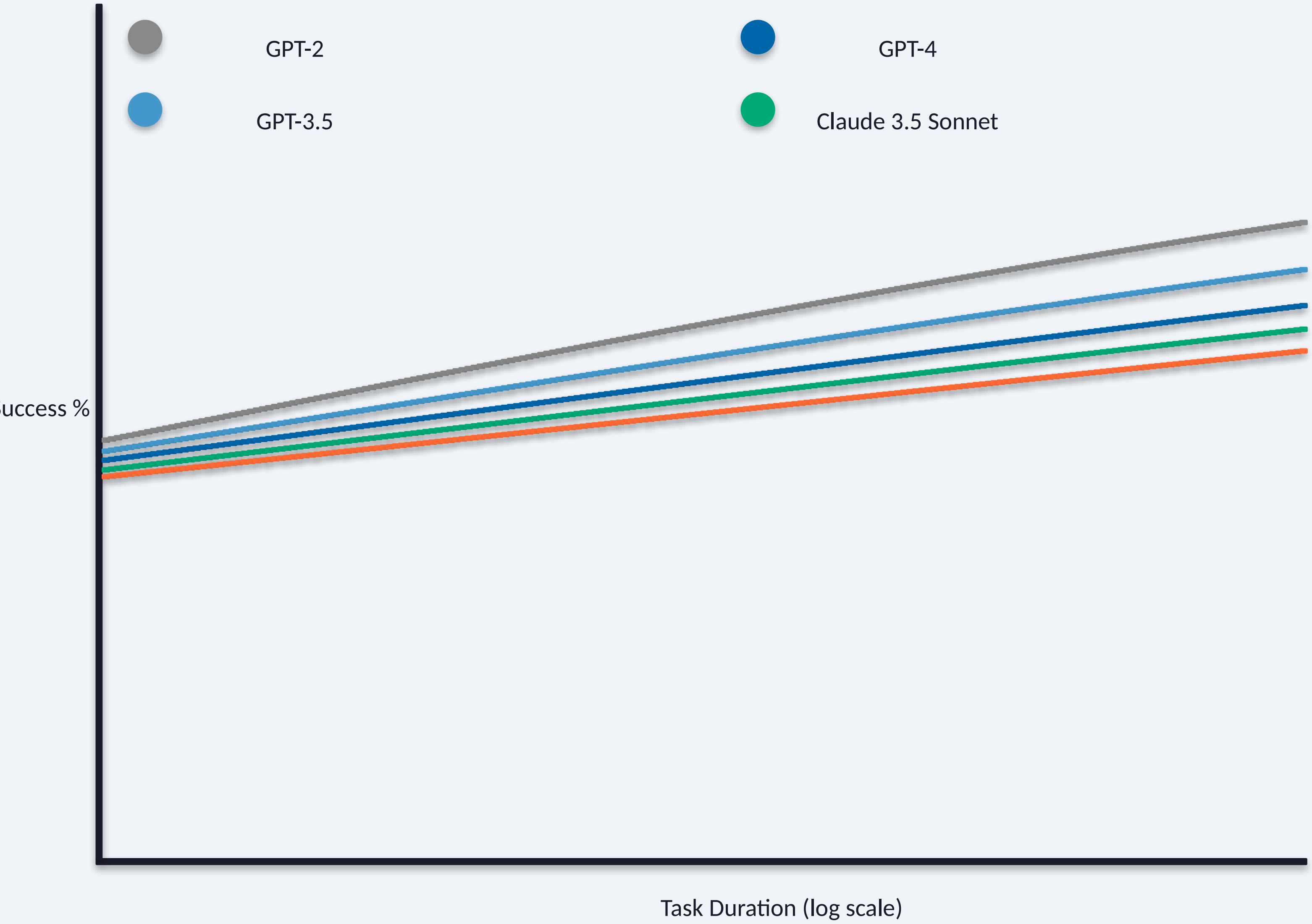


Fig. 1: Exponential trend in AI time horizon

Success Probability Drops Sharply as Task Length Increases



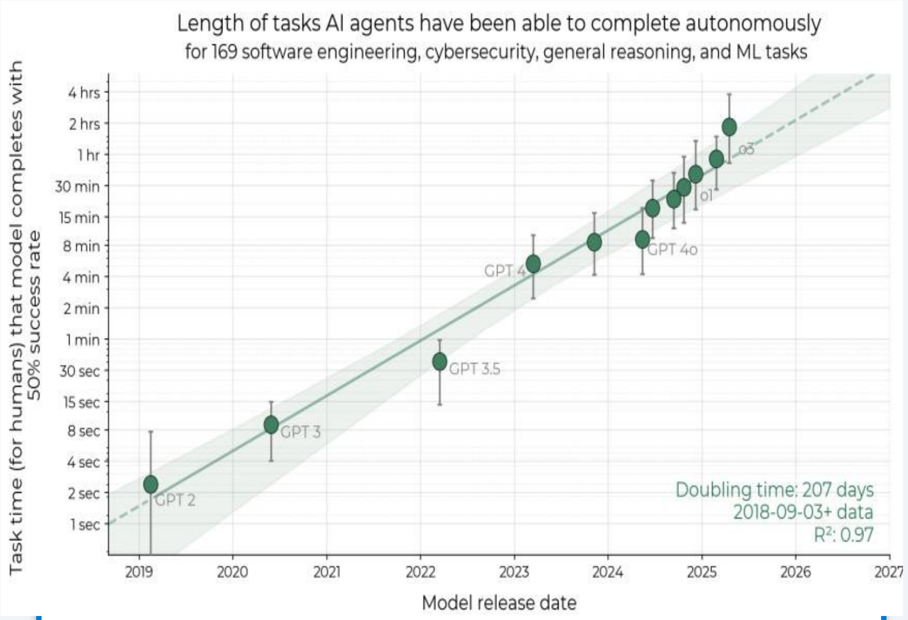


Fig. 3: Per-model success probability curves

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Improved Logical Reasoning

Enhanced chain-of-thought and multi-step reasoning capabilities allow AI to tackle more complex, longer-duration tasks

Better Tool Use

APIs, code interpreters, file system access, and web browsing enable agents to interact with real-world environments

Greater Reliability & Self-Awareness

Models better recognize when they've made mistakes, self-correct, and know when to escalate or ask for clarification

Ability to Adapt to Mistakes

Error recovery and iterative problem-solving enable agents to persist through failures rather than giving up

Limitation: Performance is notably lower on less structured, "messier" tasks that don't fit clean evaluation patterns

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Forward Extrapolation

If the trend continues, AI reaches >1 month time horizon (167 work hours) between:

Mid-2028 – Mid-2031

This implies automation of many month-long software projects

2019	2023	2025	2028-31
GPT-2 2 sec	GPT-4 5 min	o3 2 hrs	1 Month Horizon

Growth Rate Acceleration

2023–2025 horizon growth rate is ~20% faster than the overall 2019–2025 rate, suggesting possible acceleration in capability gains

Important Caveats

• External validity remains an open question — tasks may not perfectly represent real-world intellectual labor

• Trend could slow due to saturated performance on "easy" tasks or encounter new barriers

• Real-world job distributions may have different task-length profiles than benchmark suite

Takeaways: Exponential Autonomous Capability Demands Proactive Safety Governance

1. 50% time horizon is a robust, intuitive metric for tracking AI autonomy with $R^2 = 0.97$ over 6 years of model releases

2. Doubling every ~7 months with possible acceleration since 2024 — capability growth is not slowing

3. Current frontier at ~2 hours of human-equivalent work (o3) — up from 2 seconds for GPT-2 in 2019

4. External validity remains open — tasks may not perfectly represent real-world intellectual labor

5. Key limitations: poor performance on unstructured tasks; uncertain generalization to real-world job distributions

Source: METR — arXiv:2503.14499 (March 2025) | Kwa, West, Becker, et al.